

Lecture 5: Concatenating and merging

TECH2: Introduction to Programming, Data, and Information Technology

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See GitHub repository for notebooks and data:

<https://github.com/richardfoltyn/TECH2-H26>

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1 Concatenating and merging data

More often than not, data sets come from various sources and need to be concatenated (the process of appending observations or variables) or merged as part of data pre-processing. Pandas offers several routines to accomplish such tasks which we study in this unit:

1. `pd.concat()` allows us to combine multiple Series or DataFrames by appending observations (rows) or columns.
2. `pd.merge()` allows us to match observations from two DataFrames or named Series and combine these into a *merged* DataFrame.

You can also consult the official [user guide](#) and the pandas [cheat sheet](#) for more information.

1.1 Concatenation

Concatenation with `pd.concat()` is used to combine multiple data sets along the row or column axes. This function can be called with both Series and DataFrame arguments, as we illustrate below.

1.1.1 Concatenating Series

We begin with the simplest case of combining two Series to obtain a new Series which contains observations from both.

Example: Concatenating two Series along the row axis

```
[1]: import pandas as pd

# Create first series of 3 observations
```

```
a = pd.Series(['A0', 'A1', 'A2'])
a
```

```
[1]: 0    A0
      1    A1
      2    A2
      dtype: str
```

```
[2]: # Data for second series (5 observations)
      data_b = [f'B{i}' for i in range(5)]

      # Create second series
      b = pd.Series(data_b)
      b
```

```
[2]: 0    B0
      1    B1
      2    B2
      3    B3
      4    B4
      dtype: str
```

To concatenate a and b along the first dimension, we call `pd.concat()` as follows:

```
[3]: # Call concat() with the default value for axis, which is axis=0
      s = pd.concat((a, b))

      # Alternatively, make explicit that we are concatenating along the row axis
      # s = pd.concat((a, b), axis=0)
      s
```

```
[3]: 0    A0
      1    A1
      2    A2
      0    B0
      1    B1
      2    B2
      3    B3
      4    B4
      dtype: str
```

As you can see, `pd.concat()` also concatenates the index, so the index values are no longer unique. Pandas allows duplicate labels, but unintended duplicates can make selection ambiguous and prevent reindexing. We can create a unique index with the `reset_index()` method that we encountered in previous units:

```
[4]: # Reset index to get rid of duplicates
      s = s.reset_index(drop=True)
      s
```

```
[4]: 0    A0
      1    A1
      2    A2
      3    B0
      4    B1
      5    B2
      6    B3
      7    B4
      dtype: str
```

Alternatively, we can pass the `ignore_index=True` argument to `pd.concat()` and the return value will use the default index:

```
[5]: # Concatenate two series, use the default index for the return value
pd.concat((a, b), ignore_index=True)
```

```
[5]: 0    A0
      1    A1
      2    A2
      3    B0
      4    B1
      5    B2
      6    B3
      7    B4
      dtype: str
```

Example: Concatenating along the column axis

It is also possible to concatenate Series along the column dimension by specifying `axis=1`. Pandas aligns observations by index labels and inserts NaN where a label is missing from one Series, even if the Series have equal lengths.

```
[6]: s = pd.concat((a, b), axis=1)
      s
```

```
[6]:      0    1
      0    A0    B0
      1    A1    B1
      2    A2    B2
      3    NaN    B3
      4    NaN    B4
```

If the Series in question have no names, pandas assigns the values 0, 1, ... as column names. This can be avoided by explicitly passing the desired column names using the `keys` argument:

```
[7]: s = pd.concat((a, b), axis=1, keys=['Variable1', 'Variable2'])
      s
```

```
[7]: Variable1 Variable2
      0      A0      B0
      1      A1      B1
      2      A2      B2
      3      NaN      B3
      4      NaN      B4
```

Your turn

1. Create a new Series with observations ['C0', 'C1'].
2. Using the previously created Series a and b, concatenate all three objects along the row axis and create a new (unique) index.
3. Concatenate all three objects along the column axis. Assign the column names 'Column1', 'Column2', and 'Column3' using `keys`, without `ignore_index=True`.

1.1.2 Concatenating DataFrames

Concatenating DataFrames works exactly the same way as for Series.

Concatenating along the column axis

Example: Concatenating two DataFrames along the column axis

In this example, we create two DataFrames with two and three columns, respectively.

```
[8]: import numpy as np

# Create 2 x 2 array of string data
data_a = np.array([f'A{i}' for i in range(4)]).reshape((2, 2))

df_a = pd.DataFrame(data_a)
df_a
```

```
[8]:      0    1
0  A0  A1
1  A2  A3
```

```
[9]: # Create 2 x 3 array of string data
data_b = np.array([f'B{i}' for i in range(6)]).reshape((2, 3))

df_b = pd.DataFrame(data_b)
df_b
```

```
[9]:      0    1    2
0  B0  B1  B2
1  B3  B4  B5
```

To create a new DataFrame which contains the columns from both `df_a` and `df_b`, we use `pd.concat(..., axis=1)`:

```
[10]: # Concatenate along the column axis
df = pd.concat((df_a, df_b), axis=1)
df
```

```
[10]:      0    1    0    1    2
0  A0  A1  B0  B1  B2
1  A2  A3  B3  B4  B5
```

As before, the resulting DataFrame can have non-unique column names which is undesirable. There is no `reset_index()` method for columns, but we can easily create unique column names as follows:

```
[11]: # Reset column index to 0, 1, 2, ...
df.columns = np.arange(len(df.columns))
df
```

```
[11]:      0    1    2    3    4
0  A0  A1  B0  B1  B2
1  A2  A3  B3  B4  B5
```

Alternatively, we can pass the `ignore_index=True` argument to `pd.concat()` as we did earlier to have pandas create default column names.

```
[12]: # Concatenate columns, return DataFrame with default column names
pd.concat((df_a, df_b), axis=1, ignore_index=True)
```

```
[12]:      0    1    2    3    4
0  A0  A1  B0  B1  B2
1  A2  A3  B3  B4  B5
```

It is also possible to add an additional level of column names to the resulting DataFrame by specifying the `keys` argument:

```
[13]: # Concatenate along column axis, add additional column index level [A, B]
df = pd.concat((df_a, df_b), axis=1, keys=['A', 'B'])
df
```

```
[13]:      A      B
      0  1  0  1  2
0  A0  A1  B0  B1  B2
1  A2  A3  B3  B4  B5
```

The new DataFrame then has a so-called hierarchical column index.

Example: Concatenating a DataFrame and a Series

One can also concatenate DataFrames and Series objects along the column axis. The Series name becomes its column name; an unnamed Series gets a default name. Here, `ignore_index=True` assigns default names to all columns.

```
[14]: s = pd.Series(['C0', 'C1'])
s
```

```
[14]: 0    C0
      1    C1
      dtype: str
```

```
[15]: # Concatenate DataFrame and Series
pd.concat((df_a, s), axis=1, ignore_index=True)
```

```
[15]:      0  1  2
0  A0  A1  C0
1  A2  A3  C1
```

Concatenating along the row axis

We usually concatenate DataFrames along the row axis if we have observations on the same variables scattered across multiple data sets. Appending DataFrames with different columns will usually create NaN values and hence is often not useful.

Example: Concatenating rows with identical columns

```
[16]: # Concatenate 2x2 DataFrame and 3x2 DataFrame (note the transpose!)
df = pd.concat((df_a, df_b.T), axis=0, ignore_index=True)
df
```

```
[16]:      0  1
0  A0  A1
1  A2  A3
2  B0  B3
3  B1  B4
4  B2  B5
```

Example: Concatenating rows with different columns

The DataFrames `df_a` and `df_b` have a different number of columns, so the resulting DataFrame will contain NaN for all observations of column 2 that were originally in `df_a`:

```
[17]: # Concatenate DataFrame rows with different numbers of columns
df = pd.concat((df_a, df_b), axis=0, ignore_index=True)
df
```

```
[17]:
```

	0	1	2
0	A0	A1	NaN
1	A2	A3	NaN
2	B0	B1	B2
3	B3	B4	B5

Your turn

Use the data files located in the folder `../data/FRED` to perform the following tasks:

1. Load the data in `FRED_monthly_1950.csv` and `FRED_monthly_1960.csv` into two different DataFrames. The files contain monthly macroeconomic time series for the 1950s and 1960s, respectively. *Hint:* Use `pd.read_csv(..., parse_dates=[ 'DATE' ])` to automatically parse strings stored in the DATE column as dates.
2. Concatenate these DataFrames along the row dimension to get a total of 240 observations.
3. Set the column DATE as index for the newly created DataFrame.

1.2 Merging and joining data sets

1.2.1 Types of merges

While concatenation simply appends blocks of rows or columns from multiple data sets, merging allows for more fine-grained control over how data should be combined. The most common scenarios in empirical work are:

1. *one-to-one*: The observations in data sets A and B have a unique identifier (“key”), and each observation in A is matched with at most one observation in B. For example, we could have data on individuals from multiple sources, and each of these data sets identifies individuals by their social security number.
2. *many-to-one*: Data set A can contain multiple observations with the same key, while keys in data set B are unique. For example, we could have data on individuals in A and data at the ZIP-code (neighborhood) level in B. Many individuals can live in the same neighborhood, so multiple observations in A can match one observation in B.
3. *many-to-many*: Identifying keys are not unique in either data set. For each matching key, the result contains the Cartesian product of the corresponding rows from both data sets. This situation should usually be *avoided* as it tends to have surprising results and can potentially consume large amounts of memory.

1.2.2 Implementation in pandas

Merging in pandas can be performed in two different ways:

1. `pd.merge()` is a function that takes the *two* DataFrames to be merged as arguments:


```
result = pd.merge(df_A, df_B)
```
2. `df.merge()` is a method of a specific DataFrame object, and takes as an argument the other DataFrame to be merged:


```
result = df_A.merge(df_B)
```

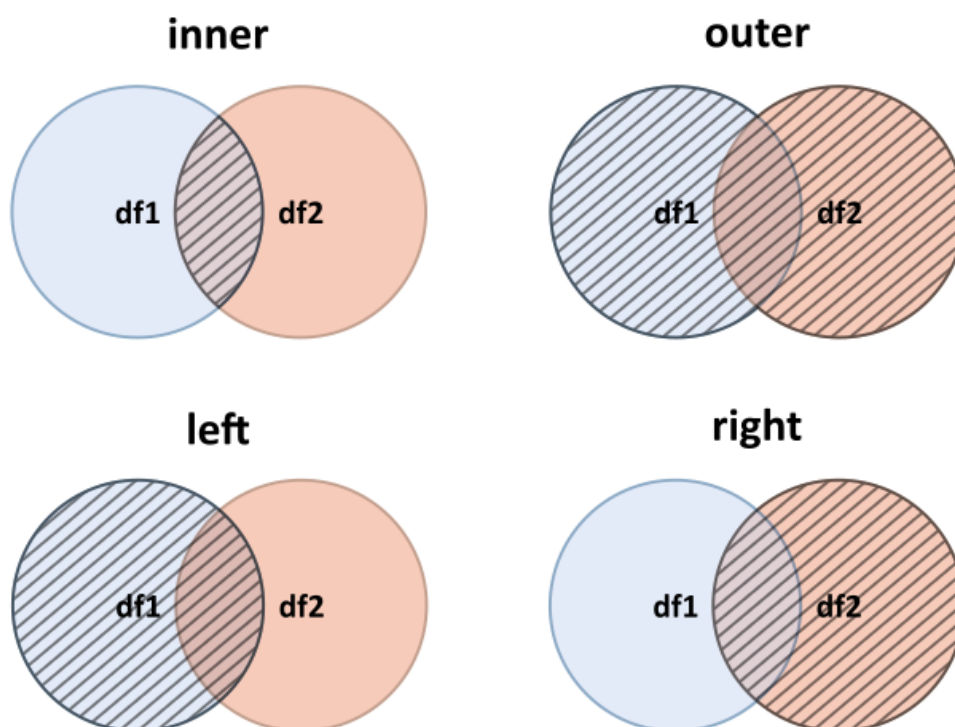
Both ways are equivalent and can be used interchangeably.

1.2.3 Controlling the resulting data set

Irrespective of whether we perform a *one-to-one* or a *many-to-one* merge, we frequently face the situation that some observations are present in one data set but not the other. We therefore need to control which subset of the data we want to retain in the final data set. This is accomplished using the `how` argument passed to `merge()`. There are several possible merge methods which were originally introduced in SQL, a data processing language for relational databases (see also the official [user guide](#)):

1. `how='inner'` performs a so-called *inner join*: the merged data contains only the *intersection* of keys that are present in *both* data sets.
2. `how='outer'` performs an *outer join*: the merged data contains the *union* of keys present in either of the data sets. Rows which are not present in both data sets will contain missing values.
3. `how='left'` performs a *left join*: all identifiers from the *left* data set are present in the merge result, but rows that are only present in the *right* data set are dropped.
4. `how='right'` performs a *right join*: all identifiers from the *right* data set are present in the merge result, but rows that are only present in the *left* data set are dropped.

The following figure illustrates these concepts graphically using Venn diagrams. Each circle represents the keys present in the left (`df1`) or right (`df2`) DataFrames. The merge method controls which subset of keys is retained in the merge result.



1.2.4 Merging with `pd.merge()`

One-to-one merges

We first create two data sets A and B used to demonstrate various merge methods. We use the column `key` as the identifier on which to perform merges.

```
[18]: # Create first DataFrame with 2 rows
df_a = pd.DataFrame({'key': [0, 1], 'value_a': ['A0', 'A1']})
df_a
```

```
[18]:   key value_a
0    0      A0
1    1      A1
```

```
[19]: # Create second DataFrame with 2 rows
df_b = pd.DataFrame({'key': [1, 2], 'value_b': ['B1', 'B2']})
df_b
```

```
[19]:   key value_b
0    1      B1
1    2      B2
```

When merging two DataFrames, in most cases we need to specify the columns (or index levels) on which the merge should be performed. We do this using the argument `on` when calling `pd.merge()`.

```
[20]: # Merge A and B on the identifier 'key' using an inner join
pd.merge(df_a, df_b, on='key', how='inner')
```

```
[20]:   key value_a value_b
0    1      A1      B1
```

Note that in this case we could leave the `on` argument unspecified, as then `pd.merge()` by default merges on the intersection of columns present in both DataFrames (which in this case is just the column `key`). However, for clarity it is advisable to always specify on explicitly.

Moreover, `pd.merge()` performs an inner join by default, so we could have called the function as follows to get the same result:

```
[21]: # Merge A and B on default key using default inner join
pd.merge(df_a, df_b)
```

```
[21]:   key value_a value_b
0    1      A1      B1
```

Since we are performing an inner join, the merged data set contains only a single row corresponding to the identifier 1, the only one present in both DataFrames.

If we want to retain all observations, we achieve this using an outer join:

```
[22]: # Merge A and B using outer join (keep union of observations)
pd.merge(df_a, df_b, on='key', how='outer')
```

```
[22]:   key value_a value_b
0    0      A0      NaN
1    1      A1      B1
2    2      NaN      B2
```

Since the keys 0 and 2 are not present in both DataFrames, the corresponding columns contain missing values.

We can also only retain the keys present in the left (i.e., the first argument) or the right (i.e., the second argument) DataFrame:

```
[23]: # Merge A and B on the identifier 'key', keep left identifiers
pd.merge(df_a, df_b, on='key', how='left')
```

```
[23]:   key value_a value_b
0    0      A0      NaN
1    1      A1      B1
```

```
[24]: # Merge A and B on the identifier 'key', keep right identifiers
pd.merge(df_a, df_b, on='key', how='right')
```

```
[24]:   key value_a value_b
0    1      A1      B1
1    2     NaN      B2
```

Your turn

Use the data files located in the folder `../data/FRED` to perform the following tasks:

1. Load the data in `CPI.csv` and `GDP.csv` into two different DataFrames. The files contain monthly data for the Consumer Price Index (CPI) and quarterly data for GDP, respectively. *Hint:* Use `pd.read_csv(..., parse_dates=['DATE'])` to automatically parse strings stored in the DATE column as dates.
2. Merge the CPI and the GDP time series with `pd.merge()` without specifying the `how` argument.
3. Merge the CPI and the GDP time series with `pd.merge()` using a left join (`how='left'`).
4. Repeat the merge using an inner join (`how='inner'`). Confirm that the result matches step 2.

How do the results differ depending on the value of the `how` argument?

Many-to-one merges

To illustrate many-to-one merges, we create a DataFrame A which has non-unique values in the column `key`. We want to merge this with another DataFrame B with unique values in the column `key`. This operation is therefore a many-to-one merge.

```
[25]: # Create first DataFrame with 4 rows, non-unique key
df_a = pd.DataFrame({'key': [0, 1, 0, 1], 'value_a': ['A0', 'A1', 'A2', 'A3']})
df_a
```

```
[25]:   key value_a
0    0      A0
1    1      A1
2    0      A2
3    1      A3
```

```
[26]: # Create second DataFrame with 2 rows, unique key
df_b = pd.DataFrame({'key': [0, 1], 'value_b': ['B0', 'B1']})
df_b
```

```
[26]:   key value_b
0    0      B0
1    1      B1
```

```
[27]: # Merge A and B on the identifier 'key' using the default inner join
pd.merge(df_a, df_b, on='key')
```

```
[27]:   key value_a value_b
0    0      A0      B0
1    1      A1      B1
2    0      A2      B0
3    1      A3      B1
```

As you can see, in the resulting DataFrame the values from B are matched with multiple rows of A.

Your turn

1. Load `../data/titanic.csv` into a DataFrame named `titanic`.
2. Create a small lookup DataFrame `ports` mapping port codes to full names:

```
ports = pd.DataFrame({
    'Embarked': ['C', 'Q', 'S'],
    'PortName': ['Cherbourg', 'Queenstown', 'Southampton'],
})
```
3. Merge `titanic` and `ports` on `Embarked` using the `pd.merge()` function with `how='left'` to retain all passengers.
4. Inspect the first few rows of the merged DataFrame. Notice how each full port name is matched to multiple passenger rows, as this is a many-to-one merge.

1.2.5 Merging with `DataFrame.merge()`

As mentioned above, there is an alternative but equivalent way to merge DataFrames using the method `df.merge()`. In this context, the *left* DataFrame is the one on which `merge()` is being invoked, while the *right* DataFrame is the argument passed to `merge()`.

To illustrate, we recreate the DataFrames we used for the one-to-one merge examples above:

```
[28]: # Create first DataFrame with 2 rows
df_a = pd.DataFrame({'key': [0, 1], 'value_a': ['A0', 'A1']})

# Create second DataFrame with 2 rows
df_b = pd.DataFrame({'key': [1, 2], 'value_b': ['B1', 'B2']})
```

We now use the `merge()` method invoked on DataFrame A to merge it with DataFrame B:

```
[29]: # Use DataFrame method to merge, keep only left identifiers
df_a.merge(df_b, on='key', how='left')
```

```
[29]:   key  value_a  value_b
0    0      A0      NaN
1    1      A1      B1
```

```
[30]: # Now df_a is the right DataFrame, set of final identifiers is the same as
# in the example above!
df_b.merge(df_a, on='key', how='right')
```

```
[30]:   key  value_b  value_a
0    0      NaN      A0
1    1      B1      A1
```

Example: Merging with overlapping column names

Sometimes both DataFrames contain the same column names. If these columns are not used as keys in the merge operation, pandas automatically renames these columns in the resulting DataFrame to avoid naming clashes.

To illustrate, we rename the value columns to `'value'` in both DataFrames and then perform the merge:

```
[31]: # Rename columns to common name 'value'
df_a = df_a.rename(columns={'value_a': 'value'})
df_b = df_b.rename(columns={'value_b': 'value'})
```

Note that once we have identical column names ['key', 'value'] in both DataFrames, we *must* specify the on argument to merge() as otherwise pandas by default merges on the intersection of column names in both DataFrames, i.e., in this case it merges on ['key', 'value']:

```
[32]: # Invoking merge() with default on argument has unintended consequences
df_a.merge(df_b)
```

```
[32]: Empty DataFrame
      Columns: [key, value]
      Index: []
```

The merge result is empty because we are performing an *inner join* (the default), and there are no overlapping rows that have the same values for both key and value columns. We therefore need to explicitly specify on='key' to get the desired result:

```
[33]: # Merge DataFrames with overlapping column 'value'
df_a.merge(df_b, on='key')
```

```
[33]:   key  value_x  value_y
0    1      A1      B1
```

As you can see, pandas automatically appends the suffixes '_x' and '_y' to the column from the left and right DataFrames, respectively. We can change this default behavior by explicitly specifying the suffixes to be appended:

```
[34]: df_a.merge(df_b, on='key', suffixes=('_left', '_right'))
```

```
[34]:   key  value_left  value_right
0    1          A1          B1
```

Your turn

Use the data files located in the folder ../data/FRED to compare inflation and the Federal Reserve policy interest rate:

1. Load INFLATION.csv and FEDFUNDS.csv into two DataFrames, parsing strings in the DATE column as dates.
2. Rename the data column in each DataFrame to 'rate' using the rename() method.
3. Merge the two DataFrames using the merge() method without specifying on. Notice that it returns only a few rows where the inflation rate happened to equal the policy rate.
4. Merge the DataFrames on DATE using suffixes=('_inflation', '_fedfunds').
5. Compare an inner join with a left join (how='left'). Why does the left join retain more observations?

1.2.6 Merging with join()

The DataFrame method join() is a convenience wrapper around pd.merge() with the following subtle differences:

1. join() can be called *only* directly on the DataFrame object, i.e., df.join(), while for merge we have both the pd.merge() and the df.merge() variants.
2. join() always operates on the *index* of the other DataFrame, whereas merge() is more flexible and can operate on either the index or on columns.
3. join() by default performs a left join, whereas merge() performs an inner join.

As a rule of thumb, you should use `join()` if you want to join DataFrames which have a similar index.

Example: joining DataFrames

We first create two DataFrames to be joined. This time, we explicitly set an index for each of them which will be used to perform the `join()`.

```
[35]: # Create first DataFrame with 2 rows
df_a = pd.DataFrame(['A0', 'A1'], columns=['value_a'], index=[0, 1])
df_a
```

```
[35]: value_a
0      A0
1      A1
```

```
[36]: # Create second DataFrame with 2 rows
df_b = pd.DataFrame(['B1', 'B2'], columns=['value_b'], index=[1, 2])
df_b
```

```
[36]: value_b
1      B1
2      B2
```

```
[37]: # Perform left join (the default option)
df_a.join(df_b)
```

```
[37]: value_a value_b
0      A0      NaN
1      A1      B1
```

```
[38]: # Join with explicit inner join
df_a.join(df_b, how='inner')
```

```
[38]: value_a value_b
1      A1      B1
```

```
[39]: # Perform an outer join
df_a.join(df_b, how='outer')
```

```
[39]: value_a value_b
0      A0      NaN
1      A1      B1
2      NaN      B2
```

Your turn

Use the data files located in the folder `../..data/FRED` to perform the following tasks:

1. Load the data in `CPI.csv` and `GDP.csv` into two different DataFrames. The files contain monthly data for the Consumer Price Index (CPI) and quarterly data for GDP, respectively. *Hint:* Use `pd.read_csv(..., parse_dates=['DATE'])` to automatically parse strings stored in the `DATE` column as dates.
2. Set the `DATE` column as the index for each of the two DataFrames.
3. Merge the CPI with the GDP time series with `join()`. Do this with both a left and an inner join.

1.3 Dealing with missing values

We already encountered missing values in earlier lectures. These are particularly likely to arise when merging or concatenating data if individual DataFrames lack some observations.

To illustrate, recall the example from above:

```
[40]: # Create two DataFrames with partially overlapping keys
df_a = pd.DataFrame({'key': [0, 1], 'value_a': ['A0', 'A1']})
df_b = pd.DataFrame({'key': [1, 2], 'value_b': ['B1', 'B2']})
```

```
[41]: # Perform outer merge, keep union of keys
pd.merge(df_a, df_b, on='key', how='outer')
```

```
[41]:   key value_a value_b
0    0      A0      NaN
1    1      A1       B1
2    2      NaN       B2
```

Since the keys in DataFrames `df_a` and `df_b` were only partially overlapping, the resulting DataFrame has missing values by construction. In what follows, we explore strategies on how to handle these missing data.

1.3.1 Dropping missing values

One strategy is to drop missing values outright, even though we might lose information that could be useful to perform data analysis if only some but not all columns contain missing values, as is the case above.

Missing values can be dropped by either

1. Using `dropna()`, potentially restricted to a subset of columns.
2. Selecting a subset of observations to keep with a boolean operation such as `notna()` or `isna()`.
3. Avoiding the missing values in the first place, e.g., by using `merge(..., how='inner')`.

Example: Dropping missing values

Consider the merged DataFrame from above. We can drop rows with missing values with `dropna()`, which by default drops all rows with *any* missing values. Alternatively, we can specify only a subset of columns to be checked for missing values.

```
[42]: # Merge with outer join, thus creating missing values
df = pd.merge(df_a, df_b, on='key', how='outer')
```

```
[43]: # Drop any row which contains at least one missing value
df.dropna()
```

```
[43]:   key value_a value_b
1    1      A1       B1
```

```
[44]: # Drop rows which contain missing values in column 'value_a', ignore missing
# values in 'value_b'
df.dropna(subset='value_a')
```

```
[44]:   key value_a value_b
0    0      A0      NaN
1    1      A1       B1
```

Your turn

Instead of using `dropna()`, drop the missing observations using either `notna()` or `isna()`.

1. Drop all rows with any missing observations.

Hint: You can apply `notna()` or `isna()` to all columns at once and combine the results with `any()` or `all()`.

2. Drop all rows with missing observations in column `value_a`.

Example: Avoiding missing values in the first place

Of course the missing values in the example above arose only because we specified `how='outer'`. Merging with `how='inner'` drops keys which are not present in both DataFrames right away, avoiding the issue of missing values (unless these are already present in the original DataFrames):

```
[45]: # Merge using inner join, drop keys not present in both DataFrames
pd.merge(df_a, df_b, on='key', how='inner')
```

```
[45]:   key value_a value_b
0    1      A1      B1
```

1.3.2 Filling missing values

Instead of dropping data, we can impute missing values in various ways:

1. `fillna()` can be used to replace missing data with user-specified values.
2. `ffill()` and `bfill()` can be used to fill missing values forward or backward from adjacent non-missing observations.
3. `interpolate()` supports various interpolation methods such as linear interpolation based on non-missing values.

Example: Replacing missing values with `fillna()`

Consider the merged DataFrame we have created above:

```
[46]: df = pd.merge(df_a, df_b, on='key', how='outer')
df
```

```
[46]:   key value_a value_b
0    0      A0      NaN
1    1      A1      B1
2    2      NaN      B2
```

We can use `fillna()` to replace missing values with some constant.

```
[47]: # Replace ALL missing values with 'Some value'
df.fillna('Some value')
```

```
[47]:   key  value_a  value_b
0    0        A0  Some value
1    1        A1        B1
2    2  Some value        B2
```

This might not be what you want as the provided non-missing value is imposed on *all* columns. It is therefore possible to specify a different value for each column using a dictionary as an argument.

```
[48]: # Use different replacement values for columns 'value_a' and 'value_b'
df.fillna({'value_a': 'Missing A', 'value_b': 'Missing B'})
```

```
[48]:   key  value_a  value_b
0    0        A0  Missing B
1    1        A1        B1
2    2  Missing A        B2
```

Your turn

Use `../data/titanic.csv` to practice column-specific imputation:

1. Load `titanic.csv` into a DataFrame named `titanic`.
2. Use the `fillna()` method with a dictionary to replace missing values in Cabin with 'Unknown', and missing values in Age with the sample average of Age.
3. Verify that neither column contains missing values in the updated DataFrame using `notna()` combined with `all()` (e.g., `titanic[['Cabin', 'Age']].notna().all()`).

Example: forward- or backward-filling missing values

Another common imputation method is to use the previous (“forward”) or next (“backward”) non-missing value as replacement for missing data.

Continuing with the DataFrame from the previous example, we can apply these methods as follows:

```
[49]: # Forward-fill missing values from previous observation
df.ffill()
```

```
[49]:   key value_a value_b
0    0        A0      NaN
1    1        A1        B1
2    2        A1        B2
```

This inserts the value 'A1' in the 3rd row of column `value_a`, but does not do anything about the missing value in column `value_b` since there is no preceding non-missing value.

Conversely, `bfill()` does the opposite and backfills the missing value in column `value_b`:

```
[50]: df.bfill()
```

```
[50]:   key value_a value_b
0    0        A0        B1
1    1        A1        B1
2    2        NaN        B2
```

Example: linear interpolation

Consider the following Series with numerical data (interpolation only makes sense for numerical data, not strings):

```
[51]: s = pd.Series([1.0, 2.0, 3.0, np.nan, 5.0])
s
```

```
[51]: 0    1.0
1    2.0
2    3.0
3    NaN
4    5.0
dtype: float64
```

We can interpolate the missing data using `interpolate()`, for example by using linear interpolation (check the documentation for many other interpolation methods).

```
[52]: # Interpolate missing values using linear interpolation
      s.interpolate(method='linear')
```

```
[52]: 0    1.0
      1    2.0
      2    3.0
      3    4.0
      4    5.0
      dtype: float64
```

Your turn

Use the data files located in the folder `../..data/FRED` to perform the following tasks:

1. Load the data in `CPI.csv` and `GDP.csv` into two different DataFrames. The files contain monthly data for the Consumer Price Index (CPI) and quarterly data for GDP, respectively. *Hint:* Use `pd.read_csv(..., parse_dates=[ 'DATE' ])` to automatically parse strings stored in the DATE column as dates.
2. Merge the CPI with the GDP time series with `merge()` using a left join. This creates missing values in the GDP column.
3. Impute the missing GDP values using `interpolate()` and replace the missing values in column GDP. Note that the default method fills trailing missing values with the last observed value. Use `limit_area='inside'` to fill only gaps between observations.

1.4 List of functions used in this lecture

The following list contains the central functions covered in this lecture. Each function name is linked to the official API documentation which you can consult for more details regarding function arguments and return values. The [\[go to section\]](#) links to the relevant section in this notebook.

1.4.1 Concatenating (appending) data

- `pd.concat()` — append Series or DataFrames along the row or column dimension [\[go to section\]](#)
- `reset_index()` — reset (potentially non-unique) index after concatenation [\[go to section\]](#)

1.4.2 Merging data

- `pd.merge()` — merge DataFrames or named Series based on some keys [\[go to section\]](#)
- `DataFrame.merge()` — merge a given (left) DataFrame with another (right) DataFrame or named Series based on some keys [\[go to section\]](#)
- `DataFrame.join()` — join a given (left) DataFrame with another (right) DataFrame or named Series using the right object's index [\[go to section\]](#)

1.4.3 Dealing with missing values

- `notna()` — check whether (one or more) columns have non-missing values [\[go to section\]](#)
- `isna()` — check whether (one or more) columns have missing values [\[go to section\]](#)
- `dropna()` — drop rows with missing observations [\[go to section\]](#)

- `fillna()` — replace missing values with a given value [\[go to section\]](#)
- `ffill()` — fill missing values by propagating the last valid observation [\[go to section\]](#)
- `bfill()` — fill missing values by using the next valid observation to fill the gap [\[go to section\]](#)
- `interpolate()` — interpolate missing values from adjacent non-missing ones [\[go to section\]](#)